

WEEZE LAARBRUCH, Germany

WMO: 100001

Lat: 51.602N

Lon: 6.142E

Elev: 32

StdP: 100.94

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.0	-11.7	1.4	-5.3	-8.7	1.8	-3.6	12.2	9.2	10.6	8.5	2.6	50	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	30.8	20.2	28.0	19.4	26.1	18.3	21.4	27.5	20.4	25.9	19.4	24.4	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.1	22.9	18.5	13.4	22.0	17.8	12.8	21.2	62.2	27.9	58.4	25.7	55.2	24.6	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.3	-9.8	35.5	4.6	1.9	-13.1	36.9	-15.7	38.0	-18.3	39.0	-21.6	40.4		
(5)			-10.3	23.6	4.4	1.0	-13.4	24.3	-16.0	24.8	-18.5	25.4	-21.8	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	2.9	3.7	6.9	10.5	14.1	17.2	19.0	18.6	15.5	11.4	6.8	4.2		
	DBStd	6.81	4.47	3.97	3.78	3.78	3.91	3.34	3.41	3.05	2.86	3.57	3.84	4.33		
	HDD10.0	868	222	177	109	38	7	0	0	0	1	25	108	182		
	HDD18.3	2836	479	409	356	235	143	61	31	33	93	214	345	437		
	CDD10.0	1216	1	1	12	54	133	217	281	266	166	70	13	3		
	CDD18.3	142	0	0	0	1	11	27	53	41	9	1	0	0		
	CDH23.3	1250	0	0	0	16	104	241	468	344	71	6	0	0		
	CDH26.7	384	0	0	0	1	23	68	164	111	17	0	0	0		
(14) Wind	WSAvg	3.4	3.8	3.8	3.6	3.1	3.1	3.0	3.1	3.0	3.0	3.2	3.4	4.0		
(15) Precipitation	PrecAvg	791	66	62	52	43	61	69	78	81	65	66	73	75		
	PrecMax	1028	143	129	113	89	115	201	141	176	163	198	122	161		
	PrecMin	601	4	12	4	0	13	12	28	25	11	21	8	25		
	PrecStd	99	32	30	30	20	26	38	34	44	41	31	30	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.9	20.0	25.2	29.8	32.1	35.0	33.0	29.1	23.9	17.2	14.0		
		MCWB	11.1	8.9	12.4	14.7	19.0	21.5	21.2	20.7	19.2	17.2	14.6	11.6		
	2%	DB	11.2	11.8	17.0	22.2	26.1	28.2	30.9	29.8	25.2	20.2	15.1	12.8		
		MCWB	9.7	9.0	11.3	13.6	17.2	19.3	20.7	20.1	18.5	16.0	13.0	10.9		
	5%	DB	10.0	10.5	14.8	19.9	23.8	26.0	27.9	27.0	22.8	18.1	13.8	11.1		
		MCWB	8.8	8.4	10.7	12.9	16.1	18.5	19.4	19.1	17.5	15.3	12.1	9.7		
	10%	DB	8.7	9.2	12.7	17.1	21.2	23.8	25.8	24.8	20.4	16.2	12.2	9.9		
		MCWB	7.8	7.6	9.5	12.0	15.1	17.3	18.4	18.1	16.2	14.2	10.7	8.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	10.9	14.0	16.2	20.5	22.2	22.7	22.7	20.5	18.3	14.9	12.5		
		MCDB	12.6	13.1	17.3	22.3	26.8	29.4	31.2	28.7	26.1	21.8	16.6	13.7		
	2%	WB	10.2	9.7	12.4	14.6	18.7	20.6	21.3	21.1	19.3	16.7	13.3	11.2		
		MCDB	11.0	11.1	15.6	20.3	24.1	25.7	27.8	26.6	23.5	19.1	14.8	12.3		
	5%	WB	8.9	8.7	10.9	13.6	17.1	19.4	20.3	19.8	18.2	15.7	12.2	10.0		
		MCDB	9.7	10.2	14.1	18.4	21.5	24.0	26.3	25.1	21.7	17.9	13.5	11.1		
	10%	WB	7.6	7.7	9.9	12.6	15.7	18.2	19.3	18.8	17.1	14.7	11.0	8.8		
		MCDB	8.4	9.1	12.1	16.6	20.3	22.6	24.3	23.3	19.8	16.5	12.4	9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.6	7.9	9.6	9.7	9.5	9.7	9.7	8.6	7.2	5.3	4.3		
		MCDBR	5.5	7.1	11.7	13.8	12.8	13.4	13.5	13.7	11.8	9.2	5.9	4.9		
	5% WB	MCWBR	4.8	5.0	6.9	6.7	5.8	6.0	5.2	5.8	5.9	5.6	4.6	4.1		
		MCWBR	5.1	6.4	9.7	11.7	11.0	11.1	11.9	11.5	9.9	8.4	5.7	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.353	0.384	0.418	0.422	0.420	0.430	0.421	0.394	0.367	0.344	0.326		
		taud	2.369	2.337	2.258	2.198	2.226	2.248	2.225	2.273	2.338	2.397	2.375	2.344		
	Ebn at Noon		671	759	802	817	832	836	820	806	784	734	646	606		
		Edh at Noon	63	84	109	130	132	131	132	119	100	79	62	55		
(44) All-Sky Solar Radiation	RadAvg		0.73	1.40	2.54	4.09	4.95	5.32	5.05	4.29	3.14	1.84	0.86	0.56		
	RadStd		0.10	0.27	0.37	0.45	0.43	0.45	0.62	0.39	0.34	0.16	0.10	0.07		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (84 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page